NON-PALINDROMIC NUMBERS NOT DIVISIBLE BY 10 THAT DIVIDE OR ARE DIVIDED BY THEIR DIGIT REVERSAL ARE CONGRUENT TO 1, 12, 78 OR 89 MODULO 100

MATH0

Abstract. Let $R(n)$ be the decimal digit reversal of n. OEIS A071685 lists the non-palindromic numbers n, not divisible by 10, such that $n \mid R(n)$ or $R(n) \mid n$, and in 2017 H. P. Dale conjectured there that every term is congruent to 1, 12, 78 or 89 modulo 100. We give a proof that uses only the two leading and two trailing digits of n: if $R(n) = kn$, they reduce the problem to a finite check over $k < 10$ and $n \bmod 100$, whose only solutions are $(k, n \bmod 100) = (4, 78)$ and $(9, 89)$, and the case $n = k\,R(n)$ follows by applying this to $R(n)$. The statement is a known result: it follows at once from the classification of base-10 reverse multiples due to Hoey (1992) and Wilson (1997), which was recorded in the OEIS before the conjecture was posed, so our contribution is the first formal proof of a known result. The proof is verified in Lean 4 against a statement that math0 formalized from the OEIS entry; its definitions are checked by the kernel against the 16 listed terms below 10^7, and it was reviewed for fidelity and frozen before the proof attempt.

1. Introduction

For $n \in \mathbb{N}$ let $R(n)$ be the number obtained by writing the decimal digits of n in reverse order, leading zeros being dropped (OEIS A004086 [15]); thus $R(1089) = 9801$ and $R(120) = 21$. The two smallest numbers that are proper divisors of their reversal, 1089 and 2178, with $9 \cdot 1089 = 9801$ and $4 \cdot 2178 = 8712$, are classical [8]. The OEIS entry A071685 [19], contributed by Labos Elemer in 2002, lists the *non-palindromic numbers n, not divisible by* 10*, such that $n \mid R(n)$ or $R(n) \mid n$*:

$$1089,\ 2178,\ 8712,\ 9801,\ 10989,\ 21978,\ 87912,\ 98901,\ 109989,\ 219978,\ \ldots$$

On December 13, 2017, Harvey P. Dale added to the entry the conjecture that every term is congruent to 1, 12, 78 or 89 modulo 100. We prove this conjecture (Theorem 2).

Prior work. Numbers n with $R(n) = kn$, $k \geq 2$, are called *reverse multiples*, and the multiples $N = k\,R(N)$ *palintiples*. Their systematic study in arbitrary bases began with Sutcliffe [9]; see Sloane [8] for a history. In base 10, Klosinski and Smolarski [5] showed that $R(n) = kn$ forces $k \in \{1, 4, 9\}$ (as recorded by D. W. Wilson in A008919 [17]). D. J. Hoey [2], in a rec.puzzles article of May 1992, described all base-10 solutions of $N = C \cdot R(N)$ with $C > 1$ by finite-state machines: only $C \in \{4, 9\}$ occurs, and N has the shape $87\cdots12$ or $98\cdots01$. Accordingly the OEIS entry A031877 [18] states (M. F. Hasler, August 18, 2014) that "All terms are of the form 87...12 = 4*21...78 or 98...01 = 9*10...89", annotated "This was proved by Hoey, 1992" by N. J. A. Sloane (October 19, 2014). D. W. Wilson stated in A001232 [13] (December 15, 1997) that the positive solutions of $9n = R(n)$ have the form $99m$ with m a palindrome with digits in $\{0, 1\}$ and no isolated digit, and in A008918 [16] that the positive solutions of $4n = R(n)$ are their doubles. Young [11, 12] introduced trees for k-reverse multiples, Sloane [8] turned them into "Young graphs" and enumerated the solutions, and Kendrick [4], Holt [3] and Webster and Williams [10] gave further accounts and generalizations.

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Dale's conjecture is an immediate consequence of Hoey's or Wilson's description. By its own comments, A071685 is the union of A008919 (the n with $R(n) = kn$, $k \geq 2$) and A031877 (their reversals). In Wilson's description m begins and therefore ends with the block 11, so the solutions of $R(n) = 9n$ and $R(n) = 4n$ are $\equiv 99 \cdot 11 \equiv 89$ and $\equiv 198 \cdot 11 \equiv 78 \pmod{100}$, and their reversals $9n$ and $4n$ are $\equiv 1$ and $\equiv 12$. The entry A071685 (revision #39 of November 5, 2025) still lists the statement as a conjecture, and the math0 referees found no publication that addresses Dale's comment explicitly. Thus the result is known; to our knowledge the present proof is its first formalization: the referees found no earlier Lean or Mathlib formalization, and the statement is not in the formal-conjectures repository [1] (checked at commit `c77fdd4`).

Our proof does not use the classification. If n has $d \geq 2$ digits and $10 \nmid n$, the two leading digits of $R(n)$ are the two trailing digits of n exchanged, and vice versa. If $R(n) = kn$, comparing both sides at the scale 10^{d-2} gives inequalities and a congruence between k, the two leading digits of n and $n \bmod 100$, and a finite check leaves only $(k, n \bmod 100) = (4, 78)$ and $(9, 89)$. This is the familiar analysis of the outer digits of a reverse multiple, carried to two digits; we claim no novelty for it.

2. Statement

The entry A071685 [19] reads in part (verbatim):

> Name: Non-palindromic numbers n, not divisible by 10, such that either n divides R(n) or R(n) divides n, where R(n) is the digit-reversal of n.
>
> Comments: The quotient R(n)/n or n/R(n) is always 4 or 9. [...]
> Conjecture: every term mod 100 is equal to 1, 12, 78, or 89. - *Harvey P. Dale*, Dec 13 2017

Here R is A004086 [15], "Read n backwards", with the comment "Leading zeros (after the reversal has taken place) are omitted"; and by a comment in A002113 [14] (Palindromes in base 10), "n is a palindrome [...] if and only if n = A004086(n)".

Definition 1. For $n \in \mathbb{N}$ let $\ell(n)$ be the number of decimal digits of n, with $\ell(0) = 0$, and let $d_0, \ldots, d_{\ell(n)-1}$ be these digits, least significant first, so that $n = \sum_i d_i 10^i$. Put $R(n) = \sum_i d_{\ell(n)-1-i} 10^i$. Call n a *term* if $R(n) \neq n$, $10 \nmid n$, and $n \mid R(n)$ or $R(n) \mid n$.

Theorem 2 (Dale's conjecture)**.** *Every term n satisfies $n \bmod 100 \in \{1, 12, 78, 89\}$.*

All four residues occur: $9801 \equiv 1$, $8712 \equiv 12$, $2178 \equiv 78$ and $1089 \equiv 89$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A071685.lean`, reviewed for fidelity, and frozen before the proof attempt (SHA-256 `0a3757cb...40c95`). Its definitions, main data check and statement are, verbatim except that docstrings, comments, the proof of `a_data` and all other declarations are omitted (the latter are `R_examples`, and the lemmas and kernel-evaluation helpers that serve the data check; five of these lemmas are reused in Section 3):

```
namespace Math0.Statements.A071685

def R (n : ℕ) : ℕ := Nat.ofDigits 10 (Nat.digits 10 n).reverse

def IsTerm (n : ℕ) : Prop := R n ≠ n ∧ ¬ 10 ∣ n ∧ (n ∣ R n ∨ R n ∣ n)

instance : DecidablePred IsTerm := fun n => by unfold IsTerm; infer_instance

noncomputable def a (n : ℕ) : ℕ := Nat.nth IsTerm (n - 1)
```

```
theorem a_data : (List.range 16).map (fun i => a (i + 1)) =
    [1089, 2178, 8712, 9801, 10989, 21978, 87912, 98901, 109989, 219978, 879912, 989901,
      1099989, 2199978, 8799912, 9899901] := by
  -- (kernel scans, omitted; see the text)

def statement : Prop :=
  ∀ n : ℕ, IsTerm n → n % 100 = 1 ∨ n % 100 = 12 ∨ n % 100 = 78 ∨ n % 100 = 89

end Math0.Statements.A071685
```

Encoding. Mathlib's `Nat.digits 10 n` is the list $[d_0, \dots, d_{\ell(n)-1}]$ of decimal digits of n, least significant first (empty for $n = 0$), and `Nat.ofDigits 10 L` evaluates a list L in the same order, $\sum_i L_i 10^i$. So `R` is R of Definition 1. Trailing zeros of n become top entries of the reversed list and contribute nothing, as A004086 prescribes ($R(8380) = 838$). Non-palindromic is `R n` $\neq$ `n`, following A002113, and divisibility on $\mathbb{N}$ is $a \mid b \iff \exists c,\ b = ac$. `IsTerm` is the entry's name clause by clause, and `statement` is Theorem 2, quantified over all terms. There is no degenerate case: $n = 0$ is excluded by $10 \mid 0$, and $R(n) = 0$ only for $n = 0$. The sequence itself (offset 1) is `a`; the statement does not use it.

Checks against the OEIS. `R_examples` checks $R(1089) = 9801$, $R(8712) = 2178$, $R(12021) = 12021$, $R(8380) = 838$, $R(120) = 21$ and $R(0) = 0$; here 12021 and 8380 are the palindrome and the multiple of 10 that the entry's example names as excluded. `a_data` checks that the definitions reproduce the first 16 listed terms, which are all listed terms below 10^7. It rests on the lemma `term_iff`: below 10^7 the terms are exactly these 16 numbers. It is proved by kernel evaluation (`decide +kernel`) of helper functions written with recursors and primitive arithmetic, each proved sound for `R` and `IsTerm`: all $n < 1000$ are scanned, and for 4 to 7 digits about 150,000 candidates on arithmetic progressions, selected by one-digit facts on the leading and last digits and by $R(n) \equiv n \pmod 9$, all proved in the same file. In Python, the fidelity reviewer recomputed the terms below 10^7, and a referee checked that the 10,000 terms of the OEIS b-file are terms and satisfy the congruence.

3. Proof

We follow the Lean proof; each statement carries the names of the corresponding declarations. Besides ℓ and R we use

$$\sigma(x) = 10\,(x \bmod 10) + \lfloor x/10 \rfloor \qquad (x \in \mathbb{N}),$$

which exchanges the two digits of a number $x < 100$ (`rev2`), and for $n \geq 10$ the scale $P_n = 10^{\ell(\lfloor n/100 \rfloor)}$; by Lemma 4 below, $P_n = 10^{\ell(n)-2}$.

Lemma 3 (`R_cons`, `len_cons`, `R_lt`, `R_R`, `len_R`; statement file)**.** (a) *If $n \geq 1$, then $\ell(n) = \ell(\lfloor n/10 \rfloor) + 1$ and $R(n) = R(\lfloor n/10 \rfloor) + 10^{\ell(\lfloor n/10 \rfloor)}\,(n \bmod 10)$.*
(b) *$R(n) < 10^{\ell(n)}$ for all n.*
(c) *If $10 \nmid n$, then $\ell(R(n)) = \ell(n)$ and $R(R(n)) = n$.*

Proof. (a) For $n \geq 1$ the digit list of n is $(n \bmod 10)$ followed by the digit list of $\lfloor n/10 \rfloor$. Reversing puts $n \bmod 10$ at the end, at position $\ell(\lfloor n/10 \rfloor)$, after the reversed digit list of $\lfloor n/10 \rfloor$, whose value is $R(\lfloor n/10 \rfloor)$. (b) $R(n)$ is the value of a list of $\ell(n)$ entries, each at most 9. (c) Since $10 \nmid n$, we have $n \geq 1$ and $d_0 \neq 0$. So the reversed list consists of digits and its most significant entry d_0 is nonzero, hence it is the digit list of its value $R(n)$. Thus the digit list of $R(n)$ is the reverse of that of n, which gives $\ell(R(n)) = \ell(n)$, and reversing once more gives $R(R(n)) = n$. □

Lemma 4 (`len_two`, `P_le`)**.** *Let $n \geq 10$ and $P = P_n$. Then $\ell(n) = \ell(\lfloor n/100 \rfloor) + 2$ and $10P \leq n$.*

Proof. The first claim is Lemma 3(a) applied to n and to $\lfloor n/10 \rfloor \geq 1$, since $\lfloor \lfloor n/10 \rfloor /10 \rfloor = \lfloor n/100 \rfloor$. Let $q = \lfloor n/100 \rfloor$. If $q = 0$, then $P = 1$ and $10P \leq n$. If $q \geq 1$, then $10^{\ell(q)} \leq 10q$, as q has $\ell(q)$ digits, so $10P \leq 100q \leq n$. □

Lemma 5 (`R_div_P`)**.** *Let $n \geq 10$ and $P = P_n$. Then $\lfloor R(n)/P \rfloor = \sigma(n \bmod 100)$: the two leading digits of $R(n)$ are the two trailing digits of n, exchanged.*

Proof. Let $q = \lfloor n/100 \rfloor$, $u = n \bmod 10$ and $t = \lfloor n/10 \rfloor \bmod 10$. Lemma 3(a) for $\lfloor n/10 \rfloor$ gives $\ell(\lfloor n/10 \rfloor) = \ell(q) + 1$ and $R(\lfloor n/10 \rfloor) = R(q) + P\,t$, and then for n it gives

$$R(n) = R(q) + P\,t + 10P\,u = R(q) + P\,(10u + t).$$

By Lemma 3(b), $0 \leq R(q) < 10^{\ell(q)} = P$. Hence $\lfloor R(n)/P \rfloor = 10u+t$, and $10u+t = \sigma(n \bmod 100)$ because $(n \bmod 100) \bmod 10 = u$ and $\lfloor (n \bmod 100)/10 \rfloor = t$. □

Lemma 6 (`R_ge`, `div_P`)**.** *Let $n \geq 10$ with $10 \nmid n$, and $P = P_n$. Then $R(n) \geq 10$ and $\lfloor n/P \rfloor = \sigma(R(n) \bmod 100)$.*

Proof. By Lemmas 3(c) and 4, $\ell(R(n)) = \ell(n) \geq 2$, so $R(n) \geq 10$. Lemma 4 for n and for $R(n)$ gives $\ell(\lfloor R(n)/100 \rfloor) = \ell(R(n)) - 2 = \ell(n) - 2 = \ell(\lfloor n/100 \rfloor)$, so $P_{R(n)} = P$. Lemma 5 applied to $R(n)$, together with $R(R(n)) = n$, gives the claim. □

The finite step concerns the multiplier k and the residue $B = n \bmod 100$. For $k, B \in \mathbb{N}$ let $A(k,B) = \sigma(kB \bmod 100)$, and let $\mathrm{Ok}(k,B)$ be the conjunction of

(i) $B \bmod 10 \neq 0$, (ii) $k \neq 1$, (iii) $A(k,B) \geq 10$, (iv) $k\,A(k,B) \leq \sigma(B) < k\,(A(k,B)+1)$.

Lemma 7 (`Ok`, `check`)**.** *If $k < 10$, $B < 100$ and $\mathrm{Ok}(k,B)$, then $(k,B) = (4,78)$ or $(k,B) = (9,89)$.*

In Lean this is the single computational step of the proof:

```
private def Ok (k B : ℕ) : Prop := B % 10 ≠ 0 ∧ k ≠ 1 ∧ 10 ≤ rev2 (B * k % 100) ∧
    k * rev2 (B * k % 100) ≤ rev2 B ∧ rev2 B < k * (rev2 (B * k % 100) + 1)

private instance : DecidableRel Ok := fun k B => by unfold Ok; infer_instance

private theorem check :
    ∀ k < 10, ∀ B < 100, Ok k B → (k = 4 ∧ B = 78) ∨ (k = 9 ∧ B = 89) := by
  decide +kernel
```

The certificate is the decision procedure itself: the `Decidable` instance for the bounded quantifiers $\forall k < 10$, $\forall B < 100$ and for `Ok` yields a Boolean, and `decide +kernel` asks the kernel to confirm by definitional unfolding (with its built-in arithmetic on numerals) that it evaluates to `true` on all 1000 pairs. No compiled code is trusted and no axiom is added (unlike `native_decide`, which adds `Lean.ofReduceBool`). For the reader we also verify the lemma by hand.

Proof of Lemma 7 by hand. Write $B = 10\beta + b$ with digits β, b, so $\sigma(B) = 10b + \beta$ and $b \geq 1$ by (i). Write $A = A(k,B) = 10a + a'$ with $a = kb \bmod 10$ and $a' = \lfloor kB/10 \rfloor \bmod 10$. By (iii), $a \geq 1$, so $k \neq 0$. By (iv), $10ka \leq kA \leq \sigma(B) \leq 10b + 9$, hence $ka \leq b$.

Last digit and leading digit. We have $2 \leq k \leq ka \leq b \leq 9$ and $kb \equiv a \pmod{10}$ with $a \geq 1$. If $k \geq 5$, then $a = 1$ and $kb \equiv 1 \pmod{10}$, so $k \in \{7,9\}$ with $b = 3$ or $b = 9$ respectively, and $k \leq b$ leaves $(k,a,b) = (9,1,9)$. If $k = 4$, then $a \leq 2$ and a is even, so $a = 2$, $4b \equiv 2$ gives $b \in \{3,8\}$, and $b \geq 8$ leaves $(4,2,8)$. If $k = 3$, then $a \leq 3$, and $3b \equiv a$ gives $(a,b) \in \{(1,7),(2,4),(3,1)\}$,

of which only $(1,7)$ satisfies $3a \le b$. If $k=2$, then $a \le 4$ is even, so $a \in \{2,4\}$; $a=2$ gives $b \in \{1,6\}$, and $b \ge 4$ leaves $b=6$, while $a=4$ gives $b \in \{2,7\}$, both < 8. So

$$(k,a,b) \in \{(2,2,6),\ (3,1,7),\ (4,2,8),\ (9,1,9)\}.$$

Second digits. Condition (iv) reads $10ka + ka' \le 10b + \beta < 10ka + ka' + k$, with $a', \beta \le 9$. For $(2,2,6)$ it requires $60 + \beta < 42 + 2a' \le 60$, which is impossible; for $(3,1,7)$ it requires $70+\beta < 33+3a' \le 60$, also impossible. For $(4,2,8)$ it says $4a' \le \beta < 4a'+4$, that is $a' = \lfloor \beta/4 \rfloor$, while $4B = 40\beta + 32$ gives $a' = (4\beta+3) \bmod 10$. For $\beta = 0,\dots,9$ the first expression takes the values $0,0,0,0,1,1,1,1,2,2$ and the second $3,7,1,5,9,3,7,1,5,9$; they agree only for $\beta = 7$. So $B = 78$, and indeed $4 \cdot 78 = 312$, $A = \sigma(12) = 21$, $84 \le 87 < 88$. For $(9,1,9)$ it says $9a' \le \beta < 9a'+9$, that is $a' = \lfloor \beta/9 \rfloor$, while $9B = 90\beta + 81$ gives $a' = (9\beta+8) \bmod 10$. The first expression is 0 for $\beta \le 8$ and 1 for $\beta = 9$; the second is $8-\beta$ for $\beta \le 8$ and 9 for $\beta = 9$. They agree only for $\beta = 8$. So $B = 89$, and indeed $9 \cdot 89 = 801$, $A = \sigma(1) = 10$, $90 \le 98 < 99$. □

Proposition 8 (`fwd`)**.** *Let $n, k \in \mathbb{N}$ with $10 \nmid n$, $R(n) = nk$ and $R(n) \ne n$. Then $(k,\ n \bmod 100)$ is $(4,78)$ or $(9,89)$.*

Proof. First, $n \ge 10$. Indeed $n \ne 0$ since $10 \mid 0$, and if $1 \le n \le 9$, Lemma 3(a) gives $R(n) = R(0) + 10^0\, n = n$, contrary to $R(n) \ne n$.

Let $P = P_n$, $A = \lfloor n/P \rfloor$ and $B = n \bmod 100$. By Lemma 6, $A = \sigma(R(n) \bmod 100) = \sigma(nk \bmod 100) = \sigma(Bk \bmod 100) = A(k,B)$. By Lemma 5, $\lfloor nk/P \rfloor = \lfloor R(n)/P \rfloor = \sigma(B)$. By Lemma 4, $A \ge 10$, which is (iii). Since $nk = R(n) \ge 10$ by Lemma 6, $k \ge 1$, and $k \ne 1$ because $R(n) \ne n$, which is (ii). Division with remainder gives

$$AP \le n < (A+1)P, \qquad \sigma(B)\,P \le nk < (\sigma(B)+1)P.$$

Hence $kAP \le kn < (\sigma(B)+1)P$, so $kA \le \sigma(B)$; and $\sigma(B)P \le kn < k(A+1)P$ (using $k \ge 1$), so $\sigma(B) < k(A+1)$. This is (iv). Moreover $10k \le kA \le \sigma(B) \le 99$, so $k < 10$, and $B \bmod 10 = n \bmod 10 \ne 0$, which is (i). Since $B < 100$, Lemma 7 applies. □

Proof of Theorem 2 (`result`). Let n be a term; then $n \bmod 10 \ne 0$.

Case $n \mid R(n)$, say $R(n) = nk$. By Proposition 8, $n \bmod 100 \in \{78, 89\}$.

Case $R(n) \mid n$, say $n = R(n)\,k$. Let $m = R(n)$. If $10 \mid m$, then $10 \mid mk = n$, which is false; so $10 \nmid m$. By Lemma 3(c), $R(m) = R(R(n)) = n = mk$, and $R(m) = n \ne R(n) = m$. Proposition 8 applied to m gives $(k, m \bmod 100) = (4,78)$ or $(9,89)$. Then $n \bmod 100 = (m \bmod 100)\,k \bmod 100$ is $312 \bmod 100 = 12$ or $801 \bmod 100 = 1$. □

Remark 9. Proposition 8 (`fwd`) contains the base-10 multiplier theorem: if $R(n) = kn$, $R(n) \ne n$ and $10 \nmid n$, then $k \in \{4,9\}$ [5, 2]. Its proof also shows that the two leading digits $\lfloor n/P_n \rfloor = A(k, n \bmod 100)$ of n are then $A(4,78) = 21$ or $A(9,89) = 10$, as in 2178 and 1089; this consequence is not stated in Lean. The full classification of the terms is not formalized here.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A071685.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A071685.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. In the namespace `Math0.Frontier.A071685` it contains the private definitions `rev2` and `Ok`, a private `DecidableRel` instance for `Ok`, and the private lemmas `len_two`, `R_div_P`, `R_ge`, `div_P`, `P_le`, `check` and `fwd`, which are the objects of Section 3 under the names given there. It also uses the five lemmas of Lemma 3, which are theorems of the statement file, not assumptions. The main declaration is `Math0.Frontier.A071685.result`, of

type `Math0.Statements.A071685.statement`; its proof is the two cases of the proof of Theorem 2, and the file ends with `#math0_settles` applied to these two names. The proof also uses Mathlib facts on `Nat.digits` and division and the tactics `omega`, `ring`, `positivity`, `norm_num` and `simp`. The only computation is `decide +kernel` in `check` (Lemma 7). The file uses no `native_decide`, no `sorry` and no `set_option`.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which rejects `sorryAx`, `Lean.ofReduceBool` and custom axioms; and (3) the type of `result` is syntactically the statement or its negation, up to metadata, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A071685.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A071685.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A071685.result' proves 'Math0.Statements.A071685.statement'
```

The data checks `a_data`, `term_iff` and `R_examples` use only these axioms too.

Toolchain and reproduction. We use Lean 4 [6] `v4.33.1` and Mathlib [7] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. Running `scripts/verify.sh` from the repository root fetches the Mathlib cache, rebuilds all math0 results with `lake build` (re-running every audit), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. For this result alone, run `lake build Math0.Statements.OEIS.A071685` (about 90 seconds on our machine, for the kernel scans of the data check) and then `lake env lean Math0/Frontier/OEIS/A071685.lean` (about 15 seconds) in `lean`.

References

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[14] The OEIS Foundation Inc., `https://oeis.org/A002113`.

[15] The OEIS Foundation Inc., `https://oeis.org/A004086`.

[16] The OEIS Foundation Inc., `https://oeis.org/A008918`; theorem of D. W. Wilson.

[17] The OEIS Foundation Inc., `https://oeis.org/A008919`; formula entry of D. W. Wilson.

[18] The OEIS Foundation Inc., `https://oeis.org/A031877`; comment of M. F. Hasler (Aug. 18, 2014).

[19] The OEIS Foundation Inc., `https://oeis.org/A071685`; conjecture of H. P. Dale (Dec. 13, 2017).

Produced by Claude (an AI model developed by Anthropic) in the math0 project